TSLA Open — Complete Research Findings

Dates: 2026-03-03 to 2026-03-05 **Data:** 5s (47 days), 1m (271 days), 5m (525 days) — IB candles **Multi-symbol:** TSLA, NVDA, AMZN, SPY (1m + 5m)

Part A: Open Scalp Strategy — DEAD END (47 days, 5s data)

Buy ATM call or put within 120s of open, TP/SL management, hard exit 9:35. Entry: wait 30s, compare bar close to 9:30 open. Close ≥ open → Call, Close < open → Put.

Best configs

Config	Trades	Win%	Total P&L	Sharpe
Puts only (TP=$2, SL=$1)	**30**	**53%**	**+$515**	**3.6**
Both directions (TP=$2, SL=$1)	44	48%	+$355	1.7
Calls only (best)	14	43%	-$85	-3.8

What we tested and ruled out

1. **13 entry triggers** → `wait_30s` best. 15s/20s = too noisy, 60s = move already done.
2. **Tight stops** get clipped by 5s noise. Only TP≥$1.50 / SL≥$0.75 works.
3. **Trailing stops** → best +$17. Fixed TP beats trailing at the open.
4. **Indicator exits** (EMA9/20, VWAP, reversal, stall) → all worse than TP/SL.
5. **Buy-the-dip** (always long, enter on 35% recovery) → all negative (-$100 to -$1312).
6. **1m close as direction** → catastrophically worse. Wait_60s: -$582 vs wait_30s: +$454.
7. **Always-call with dip recovery** → only TP=$1.00/SL=$0.25 positive (+$172, 50% win). Not tradeable.

Verdict: The puts-only edge was regime-dependent (47-day TSLA downtrend). Over 271+ days direction is 50/50. Real spreads wipe all profit.

Part B: Direction Study (271 days 1m, 525 days 5m)

Opening direction is a coin flip (except AMZN)

Symbol	TF	Days	30s Below	1m Below	5m Below	Day Close
TSLA	1m	221	49%	52%	51%	50/50
TSLA	5m	476	49%	49%	50%	49/50
NVDA	5m	476	53%	53%	49%	51/48
AMZN	**5m**	**476**	**57%**	**57%**	**55%**	**49/49**
SPY	5m	476	47%	47%	44%	52/46

AMZN has a persistent bearish opening bias (55-57% below open). SPY has mild bullish bias.

All symbols trade above open during the day

Symbol	Day High Above Open	Avg Day High
TSLA	99%	+$7.45
NVDA	96-97%	+$2.25
AMZN	97-98%	+$2.20
SPY	97-99%	+$2.97

Part C: Day High Timing (271 days 1m)

U-shaped distribution

Time Window	TSLA (1m)	TSLA (5m)
9:30-10:00	**41%**	**42%**
10:00-11:00	15%	16%
11:00-15:00	26%	25%
15:00-16:00	**17%**	**15%**

Bull vs bear days — completely different profiles

Metric	Bull Days (136)	Bear Days (135)
High in first 5m	2%	52%
High in first 30m	10%	73%
High in last hour	36%	0%
Median high time	12:57	9:34

Descriptive, not predictive — you only know which type of day it is after the fact.

Part D: Bull/Bear Day Prediction (271 days, 1m)

What DOESN'T predict direction

- 1st bar range, volume, day range — identical for bull/bear
- Overnight gap — useless (+6pp for gap up, -0pp for gap down)
- Day of week — zero signal

Prediction accuracy scales linearly with time

Signal	→ Bull%	n	Edge
1st bar green	56%	131	+13pp
5m above open	60%	138	+21pp
15m above open	72%	128	+42pp
30m above open	77%	127	+51pp

No early shortcut. Best combos: "30m up + 5m up" = 76% bull (n=89), "30m up >$2 + gap up" = 85% (n=47).

Part E: Level Bounce Study (271-476 days, 4 symbols)

When the opening dip (first 5m) touches a known key level (prev day, 2d-50d lookback), does it bounce better?

Cross-symbol summary

Symbol	TF	Days	Level%	Lvl 5m Win%	No-Lvl 5m Win%	Edge
TSLA	1m	221	30%	84%	81%	+3pp
NVDA	1m	221	67%	82%	75%	+7pp
AMZN	1m	322	69%	85%	83%	+2pp
SPY	1m	221	63%	90%	83%	+7pp

The real edge: $1-2 dips + level touch

Symbol	$1-2 Dip + Level Win%	$1-2 No-Level Win%	Edge
NVDA	83%	62%	**+21pp**
AMZN	78%	67%	**+12pp**
SPY	90%	71%	**+19pp**
TSLA	85%	90%	-5pp

$1-2 dip Goldilocks zone: big enough to reach a level, small enough to bounce reliably.

5-day+ lookback = sweet spot

Depth	TSLA 5m Win%	NVDA	SPY
prev_day	79%	83%	90%
5d level	**100%**	**95%**	**91%**
50d level	100% (n=2)	100% (n=12)	94% (n=32)

HIGH levels > LOW levels (consistent across symbols)

Symbol	HIGH 5m Win%	LOW 5m Win%
TSLA	94%	87%
NVDA	86%	79%
AMZN	88%	81%
SPY	90%	88%

Touching a previous high (now support) = bullish structure. Touching a previous low = bearish extending.

Part F: HOLD or BAIL — Call Holder's Playbook (271 days, 1m)

Setup: You hold TSLA calls at market open. What tells you to hold or sell?

Baseline

- Avg day high above open: **$7.45** (median $5.68)
- Avg day close: **$-0.09** (coin flip)
- 89% of days reach ≥$1 above open, 55% reach ≥$5

The 5-Minute Rule

5m Signal	Days	% Bull Close	Avg Day Close	Avg Day High	30m MFE
Above open	134	**67%**	**+$3.22**	$9.99	$6.24
Below open	133	**32%**	**-$3.64**	$4.83	$2.13

$6.87 expected value gap. This is the single most powerful signal.

The Decision Matrix (best combos)

Signal	Days	% Bull	Avg Day Close	Action
5m > +$2	**76**	**75%**	**+$4.71**	**HOLD**
5m UP + big dip recovered	**73**	**67%**	**+$3.32**	**HOLD**
1m DOWN → 5m UP (reversal)	**32**	**72%**	**+$3.67**	**HOLD** (strongest!)
5m DOWN + small dip (<$1)	9	11%	-$7.70	**BAIL FAST**
5m DOWN + not recovered	97	29%	-$4.34	**BAIL**
1m UP → 5m DOWN (fakeout)	30	30%	-$3.11	**BAIL**

Recovery speed matters

Recovery After Dip	Days	% Bull	Avg Day Close

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1-2 minutes	87	55%	+$1.78
2-5 minutes	51	**67%**	**+$2.54**
Never recovered	62	**18%**	**-$7.71**

If not recovered by 5m → 82% chance of losing day. BAIL.

Bear day escape window

Even on bear days, 77% reach ≥$1 above open — but avg bear day high is at **0:29 after open**.

- By 5m: only 32% still above open
- By 30m: only 21% above open

On a bad day you have **seconds, not minutes** to exit near breakeven.

When to take profit (bull days)

- Avg bull day high: **$11.73**
- Avg time of high: **3:36 after open** (median 3:34)
- At 30m: 73% still above open, avg +$2.96

Bull days trend all day. No rush to sell in the first hour.

The Three Rules

1. **Wait until 9:35.** Price above open = HOLD (67% bull, +$3.22 avg). Below = BAIL (32% bull, -$3.64 avg).

2. **Dip-then-recovery is the STRONGEST hold signal.** 1m down → 5m up = 72% bull, +$3.67 avg. Recovery proves buying pressure.

3. **No recovery by 5m = exit.** 82% chance of losing day, avg -$7.71. Don't hope.

Files

File	Description
`open_scalp_analysis.py`	Main scalp sim (triggers, TP/SL grid, trailing, direction filter)
`open_scalp_exits.py`	Indicator-based exit comparison
`open_scalp_dip_buy.py`	Buy-the-dip strategy (always long)
`open_scalp_1m_direction.py`	30s vs 1m direction + wait time comparison
`open_scalp_call30s.py`	Always-call strategy, dip recovery + hold/momentum/TP-SL
`open_direction_study.py`	Above/below open at 30s intervals, 3 timeframes

File	Description
`day_high_timing.py`	Day high timing analysis, bull/bear split
`open_level_bounce.py`	Level bounce study (TSLA, extended lookbacks)
`open_level_multi.py`	Multi-symbol level bounce + direction (TSLA/NVDA/AMZN/SPY)
`open_hold_or_bail.py`	Call holder HOLD/BAIL decision framework
`open-scalp-results.md`	Full scalp numerical results
`open-scalp-exits.md`	Indicator exit results
`open-scalp-dip-buy.md`	Dip-buy results
`open-scalp-1m-direction.md`	30s vs 1m direction results
`open-scalp-call30s.md`	Always-call results
`open-direction-study.md`	Direction study results
`open-level-bounce.md`	TSLA level bounce results
`open-level-multi.md`	Multi-symbol level bounce + direction
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